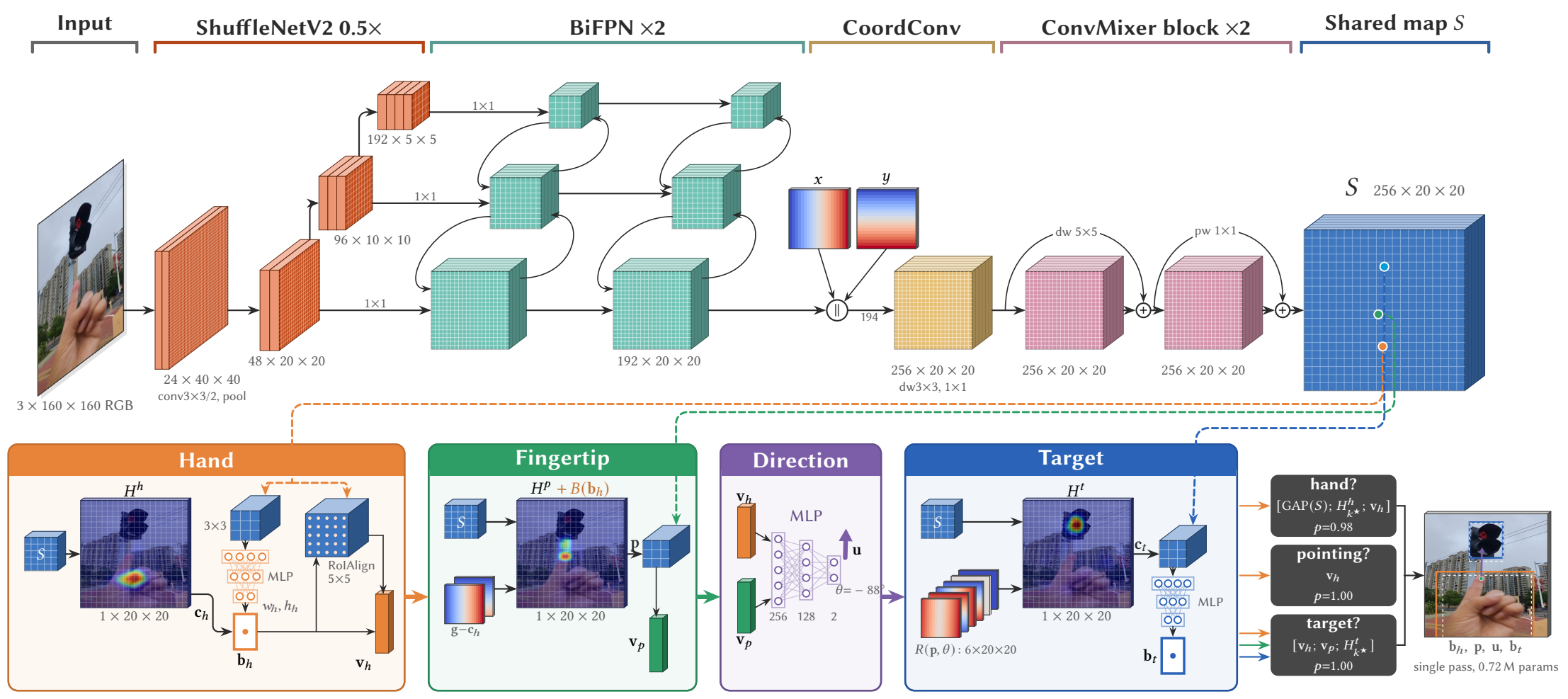

Input
ShuffleNetV2 0.5×
BiFPN ×2
CoordConv
ConvMixer block ×2
Shared map S
3 × 160 × 160 RGB
24 × 40 × 40
conv3×3/2, pool
48 × 20 × 20
96 × 10 × 10
192 × 5 × 5
1×1
192 × 20 × 20
x
y
194
256 × 20 × 20
dw3×3, 1×1
dw 5×5
pw 1×1
256 × 20 × 20
256 × 20 × 20
S 256 × 20 × 20
Hand
H^h
1 × 20 × 20
3×3
MLP
RoIAlign 5×5
w_h, h_h
$\mathbf{c}_h$
$\mathbf{b}_h$
$\mathbf{v}_h$
Fingertip
$H^p + B(\mathbf{b}_h)$
1 × 20 × 20
g−$\mathbf{c}_h$
p
$\mathbf{v}_p$
Direction
$\mathbf{v}_h$
$\mathbf{v}_p$
MLP
256 128 2
u
θ = − 88°
Target
H^t
1 × 20 × 20
$R(\mathbf{p}, \theta)$: 6×20×20
$\mathbf{c}_t$
MLP
$\mathbf{b}_t$
hand?
$[\mathrm{GAP}(S); H^h_{k^\star}; \mathbf{v}_h]$
p=0.98
pointing?
$\mathbf{v}_h$
p=1.00
target?
$[\mathbf{v}_h; \mathbf{v}_p; H^t_{k^\star}]$
p=1.00
$\mathbf{b}_h$, p, u, $\mathbf{b}_t$
single pass, 0.72 M params